Top market stocks analysis

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Stock price

- Microsoft stock
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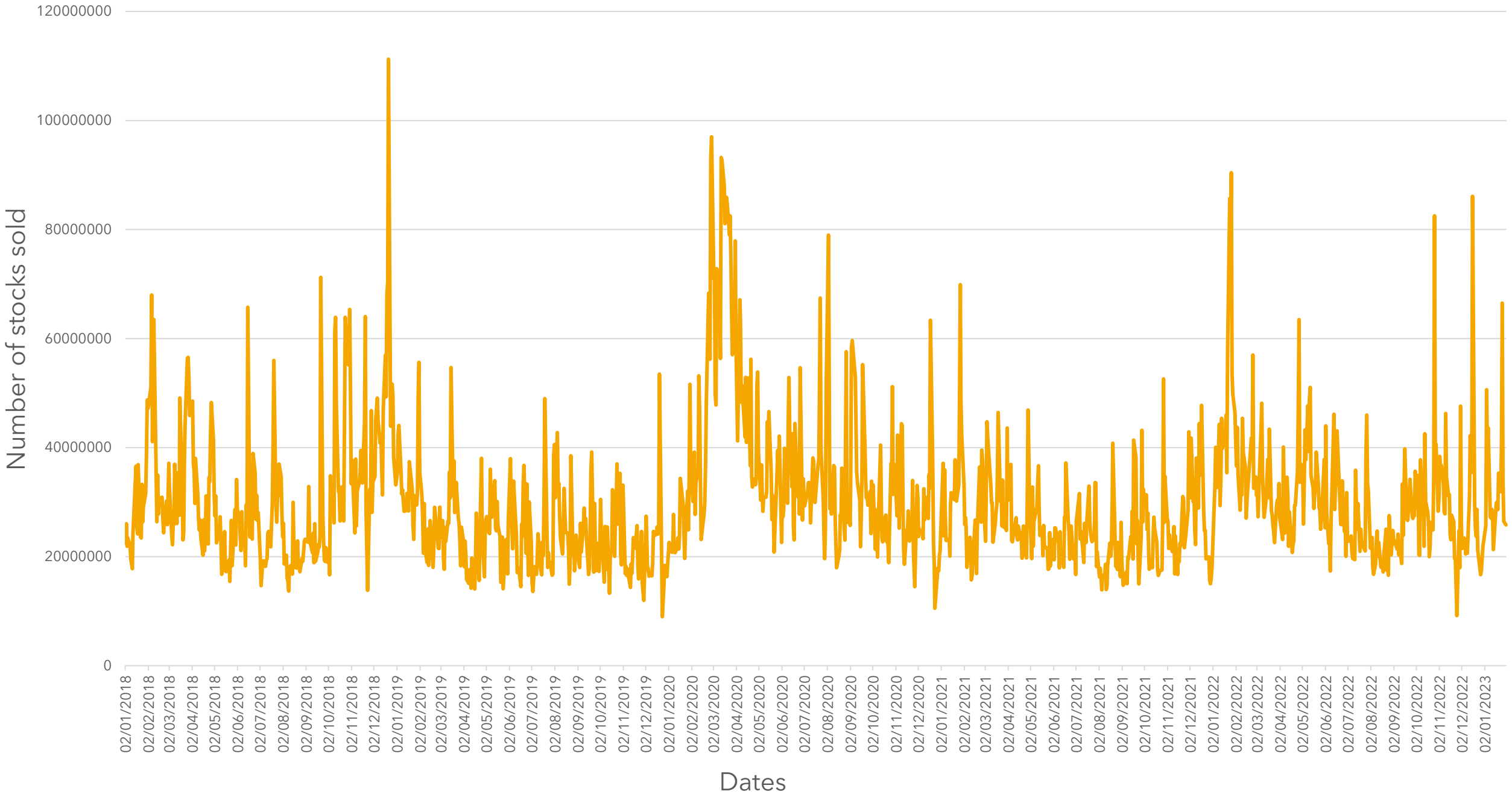
Microsoft stock

2018 -2023

Volume of Microsoft stocks sold per day (2018-2023)



Yearly variation

According to the Microsoft stock data figure showing the volume of Microsoft stocks sold per day, we can see that the stocks during this 5-year period stayed within the same range and followed the same type of fluctuating pattern around a mean of 30267789.44 sales.

Table 1- Microsoft

Price variation 2018-2023				
	2018 high	*2018 low*	*2023 high*	*2023 low*
Mean	102.1140637	99.91928312	239.2321055	233.5589464
Standard Deviation	7.905517582	7.90107238	6.979796603	7.266477396
Sample Variance	62.49720824	62.42694475	48.71756062	52.80169374
Range	29.870002	31.099998	24.070007	27.479996
Minimum	86.309998	83.830002	225.759995	219.350006
Maximum	116.18	114.93	249.830002	246.830002
Sum	25630.63	25079.74006	4545.410004	4437.619982
Count	251	251	19	19

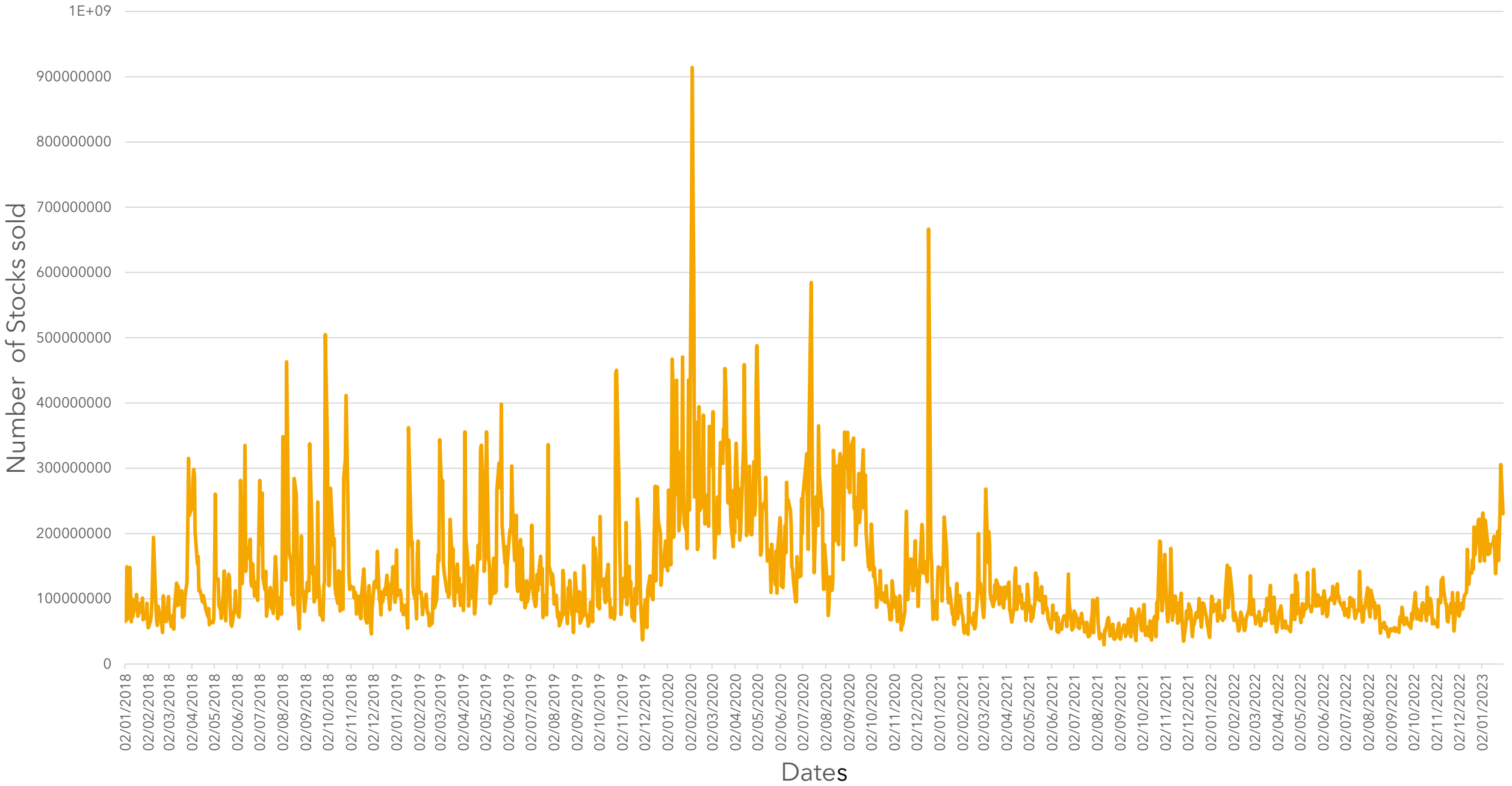
2018 vs 2023

- Looking at the data we can see that both the lowest and highest mean stock values are greater in 2023 than in 2018. Therefore, it can be inferred that there was an increase in the general Microsoft stock prices during this 6-year period.
- The data also shows that the mean stock values in 2023 had a slightly lower standard deviation and range of values than in 2018, meaning the stock is more stable now than 5 years ago and might be a more secure investment now.
- One of the most significant changes is the maximum value of the stock, which is way greater in 2023 at about 250 compared to a maximum of just about 116 in 2018. This happened even though just 19 days of data was available for 2023.

Tesla Stock

2018 -2023

Volume of tesla stock sold per day (2018-2023)



Yearly variation

- From the figure we see that the tesla stocks didn't follow a defined pattern but observed unpredictable and drastic stock sale fluctuations between 2018 and 2019.
- In 2020 a different pattern can be observed. We can see that the number of stocks sold per day is higher in 2020 compared to other years.
- Whereas from 2021 to 2023 the stock sales dropped, and we can notice that it has fewer and less significant fluctuations

Table 2- Tesla

Price variation 2018-2023				
	2018 high	*2018 low*	*2023 high*	*2023 low*
Mean	21.57957512	20.70348743	135.2926306	126.6226317
Standard Error	0.122256339	0.121312247	4.675945309	4.293561789
Standard Deviation	1.936904675	1.921947447	20.38197307	18.71520195
Range	8.475334	8.168667	68.929993	64.690002
Minimum	17.355333	16.306	111.75	101.809998
Maximum	25.830667	24.474667	180.679993	166.5
Sum	5416.473354	5196.575345	2570.559982	2405.830002
Count	251	251	19	19

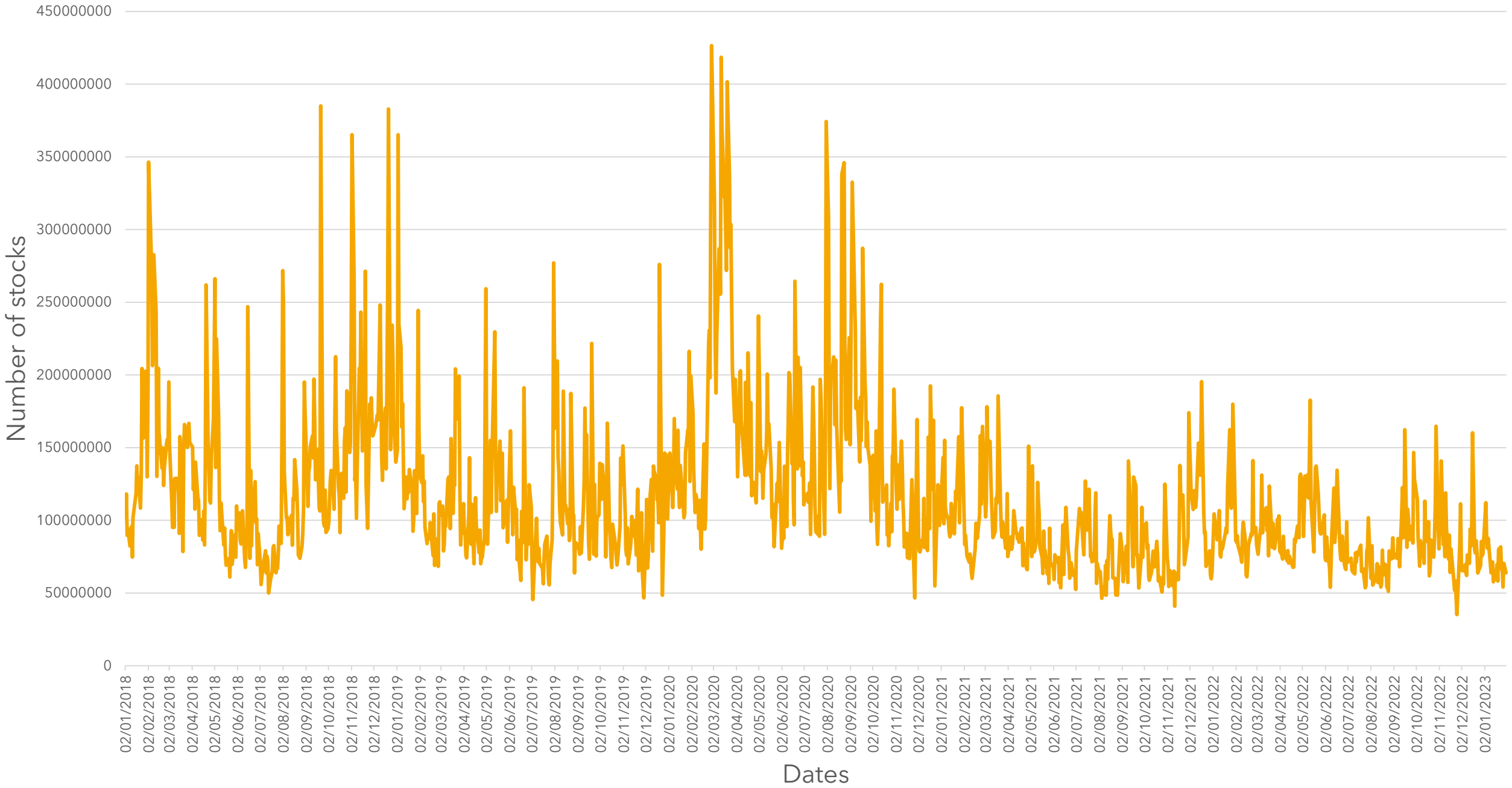
2018 vs 2023

- Just as on the Microsoft stock table suggests an overall increase in stock prices, the same applied for the tesla stock which shows an increase of about 550%.
- But unlike the Microsoft stock the range of tesla stock prices rose by about 730% and the standard deviation of its prices is also significantly greater in 2023 than in 2018, therefore suggesting that the tesla stock has become more volatile and unreliable in terms of withholding its value.

Apple Stock

2018 -2023

Volume of Apple stocks sold per day (2018-2023)



Yearly variation

- From the graph we can see that the Apple stock saw a multitude of abrupt fluctuations in its prices from 2018 right up to 2021. Showing that the demand for these stocks were volatile and weren't consistent during this period.
- On the other hand, from 2021 to 2023 the number of stock sold was more stable as it fluctuated less over a smaller range of values. This means that there was a lower but more consistent demand for apple stock.

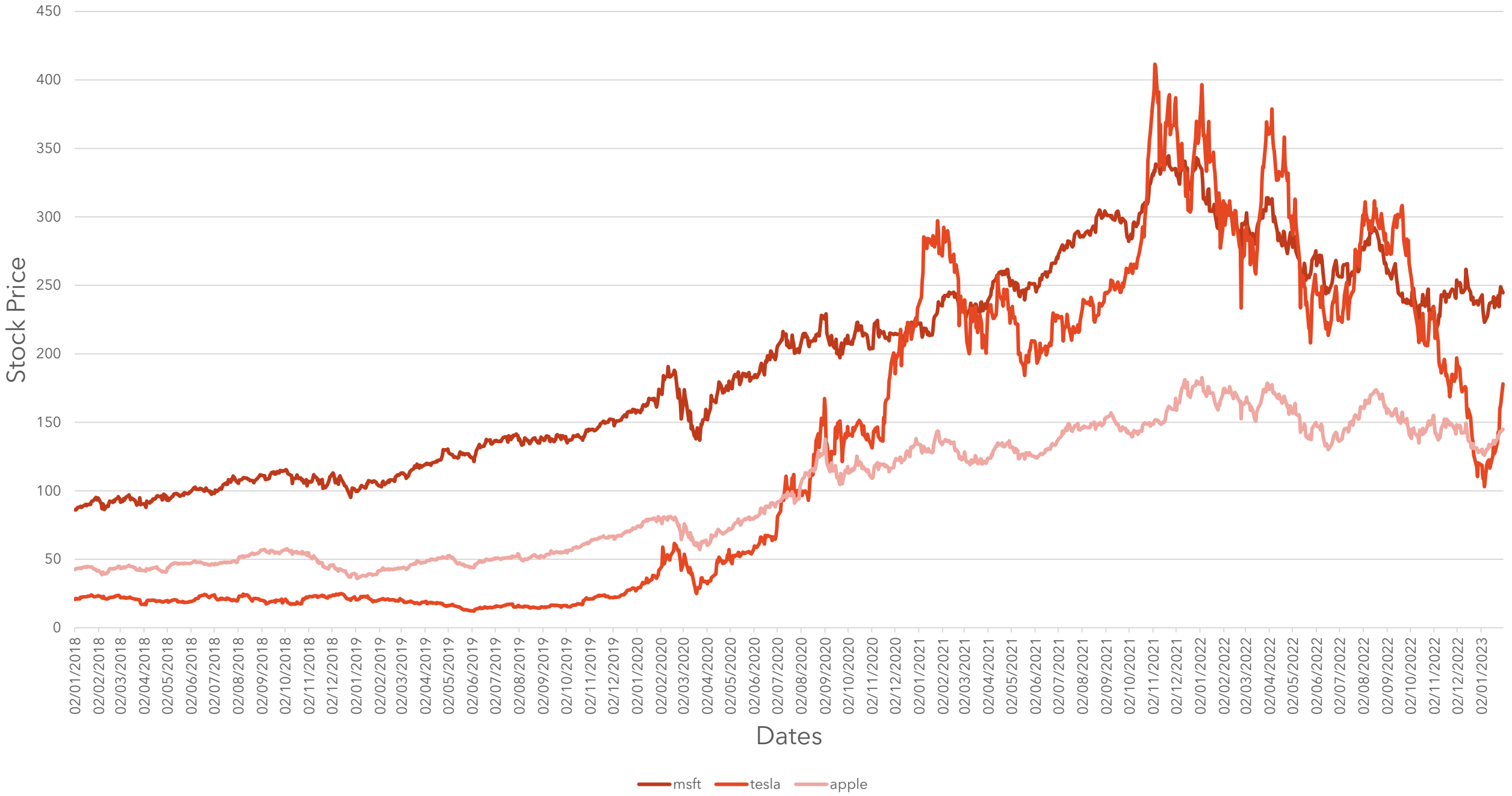
Table 3- Apple

Price variation 2018-2023				
	high 2018	*low 2018*	*high 2023*	*low 2023*
Mean	47.74852593	46.79587657	136.8994724	133.2873692
Standard Deviation	5.146255225	5.109995603	6.00282611	6.295224729
Sample Variance	26.48394284	26.11205507	36.03392131	39.62985439
Range	20.479999	20.797501	19.459999	18.910004
Minimum	37.887501	36.647499	127.769997	124.169998
Maximum	58.3675	57.445	147.229996	143.080002
Sum	11984.88001	11745.76502	2601.089976	2532.460014
Count	251	251	19	19

2018 vs 2023

- Like the Microsoft and Tesla stocks the mean price of the stock rose significantly over the 6-year period.
- But unlike the other stocks the range of the price values rose slightly showing an increase in volatility, which is further solidified by an increase in the standard deviation of the stock prices.
- The minimum and maximum price of the stock rose during this period, meaning that the stock holds more value in 2023 than it did in 2018.

Opening price of stocks

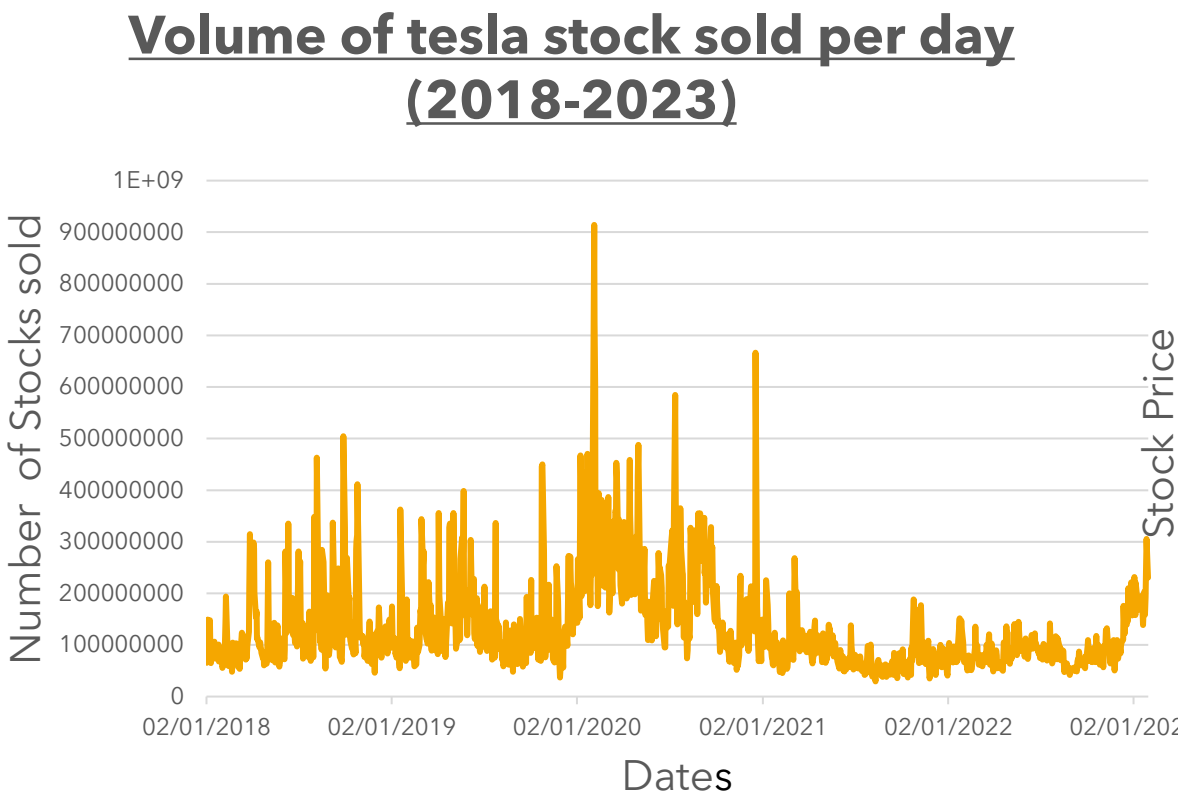


SP-500 Opening stock price (2018-2023)

Top Stock Comparison

- From the graphs we can see that the three companies follow the same pattern of growth in stock value and price as that of the sp-500 companies. Which suggests that there is an underlying factor that affects all four of these entities.
- This link is further solidified when looking at the period between march and April of 2020. Whereby all four stocks drastically fell in value at the same time. This period matches with that of the covid pandemic which suggests that the effects of the covid-19 pandemic might have had a negative consequence on stock prices.

Top Stock Comparison



Top Stock Comparison

- When looking at the above side by side comparison of the price against time and volume of stocks against time graph, we can notice that the graphs are inversely related. This may suggest that the price of the stock and the volume of the stocks sold have an inverse relationship as the rise in the stock prices take place during a decrease in stock volume sold.

Table 4- SP_500 & Apple Regression

Regression Statistics	
Multiple R	0.953257183
R Square	0.908699256
Adjusted R Square	0.908627704
Standard Error	201.4864692
Observations	1278

- The high R-squared value between the SP_500 prices and apple prices suggest a strong correlation between the top market stocks (Tesla, Microsoft & Apple) and other stocks counted in the SP_500.
- This Supports the idea that these 4 stocks are connected.

Conclusion

- The results led to three major conclusions, which are;
  - The stocks gained value and overall became more stable with time.
  - The stock prices are inversely related to the volume of stocks sold.
  - The various stocks follow a similar pattern and are connected.

Thank You